Answer **all** the questions in this Section.

1. In a simple pendulum experiment to determine the acceleration of free fall g, the following equation is used

$$T = 2\pi\sqrt{\frac{l}{g}}.$$

The following measurements were obtained using a metre rule and stopwatch respectively.

Length of the pendulum: $l = (98.0 \pm 0.1)$ cm

Average time of 10 oscillations: $t = (19.8 \pm 0.2)$ s

(a) **(i)** The calculated acceleration of free fall g is 9.869 m s^{-2}.

Determine the acceleration of free fall with its associated uncertainty.

acceleration of free fall g = ... ± ..... m s^{-2} [3]

(ii) Using a different set of apparatus, another student obtained g to be (11.15 ± 0.02) m s^{-2}.

Compare the accuracy and precision of the two sets of readings.

.....

.....

[1]

.....

(b) Tempered glass screen protector is made of many atoms.

Estimate the number of atoms in a 0.5 mm thickness tempered glass screen protector for a mobile phone. Show your working and reasoning clearly.

number of atoms = ...

[4]